



Non-salmon catch returns in the Wye catchment

A report prepared on behalf of Wye Salmon Association (WSA)

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Introduction

Whilst Salmon catches in the Wye catchment are widely reported, similar data for other species is sparse to non-existent. Likewise, electrofishing surveys by EA/NRW are limited in number, to say the least. However, a good selection of the fishing within the catchment is available on day tickets from the Wye & Usk Foundation (WUF) where catch reporting is a requirement of booking. These catch returns are freely available for viewing on their [Fishing Passport](#) website, albeit in a fairly clunky way which makes it almost impossible to analyse. To get around this, I have used a web scraping application to automate the process, and have downloaded all catch returns for trout, grayling and coarse fishing across several areas of the Wye catchment. Useful data commences from 2013-14.

Analysis

Irfon & Ithon

The quality of winter grayling fishing on these tributaries has declined immensely. Figure 1 shows the falling average catch per angler-day over the last 10 years. The regression line (grey) indicates an overall reduction by a factor of around 4.

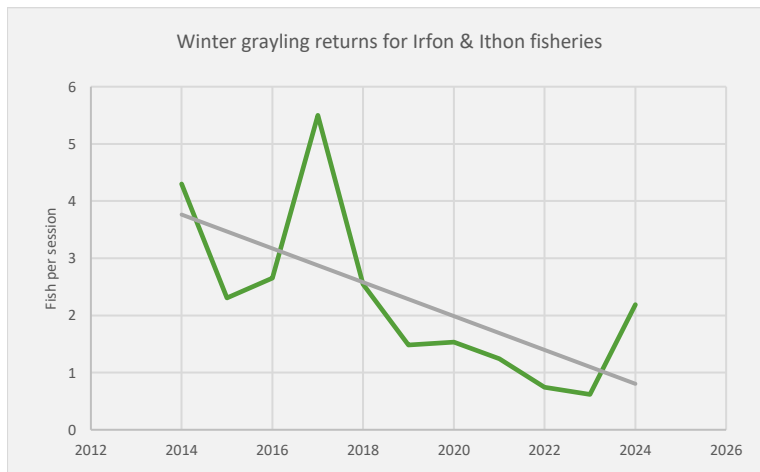


Figure 1: average winter grayling returns for all Irfon & Ithon booking office beats. The linear regression line to the data (grey) reveals the underlying trend of a four-fold fall in grayling numbers over the last decade.

Homing in on a specific Irfon fishery, figure 2 looks at Cefnllysgwynne – a beat I’ve personally visited every year since 2011. The average catch follows a very similar trajectory, with a four-fold fall since 2014. Indeed, my personal catch records suggest that the river was already in decline by 2014. Referencing 2012, my personal catches have fallen by a factor of five.

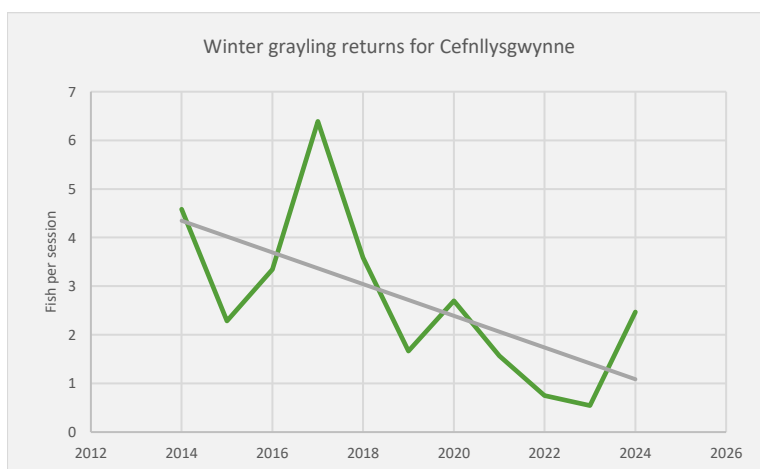


Figure 2: winter grayling returns for Cefnllysgwynne on the lower Irfon, a once prolific fishery. The linear regression line to the data (grey) reveals the same underlying trend of a four-fold fall in grayling numbers since 2014.

Increasingly, disappointed anglers have been voting with their feet. Figure 3 shows the number of angler days recorded for winter grayling on Cefnllysgwynne. Since 2011, winter angling bookings have

fallen by a factor of 7.4. The steep decline between 2011 and 2013 is strongly supportive of the suggestion that the decline in grayling numbers commenced well before 2014.

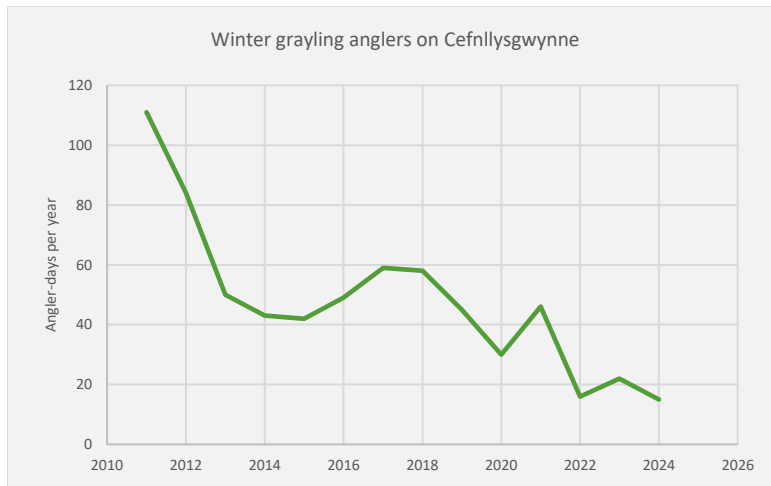


Figure 3: winter grayling angler-day numbers for Cefnllysgwynne on the lower Irfon. Bookings have fallen from 111 in 2011 to 15 in 2024.

In-season trout captures over the same period do not show a similar decline. There is considerable year to year variation for trout, but the long-term trend has been steady at around 2 fish per angler¹. Out of season trout captured by winter grayling anglers, on the other hand, have increased by a factor of three since 2014. This is most likely explained not by a population increase, which would surely also reflect in the in-season catch rate, but by the reduced abundance of grayling.

Upper Wye

Upper Wye refers to the main stem from source to Hay on Wye – essentially the Welsh part; fisheries run from around Llangurig, down to Llyswen. This part of the Welsh catchment also shows a decline in grayling catches, as illustrated by figure 4. The fall-off, roughly a factor of two, is less severe than seen on the Irfon & Ithon above.

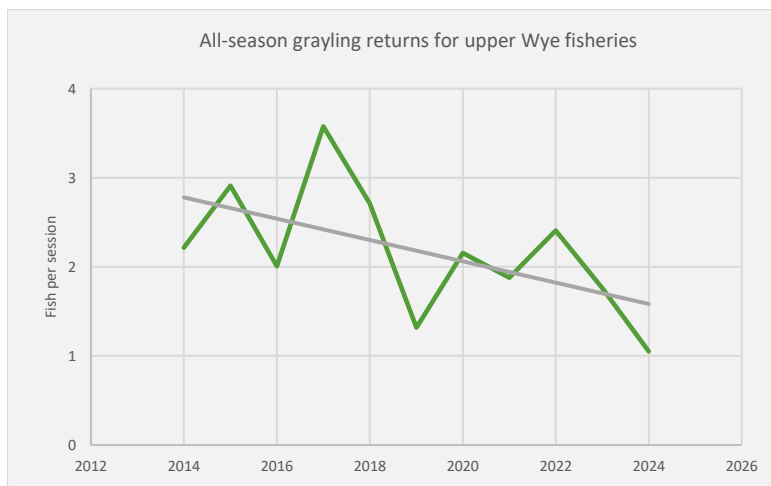


Figure 4: average (all year) grayling returns for all upper Wye booking office beats. The linear regression line to the data (grey) reveals the underlying trend of a circa two-fold fall in grayling numbers over the last decade.

Trout captures over the same period have remained substantially constant, averaging 2-3 fish per visit as shown in figure 5 overleaf.

¹ Average trout captures across all Irfon & Ithon fisheries have similarly held steady.

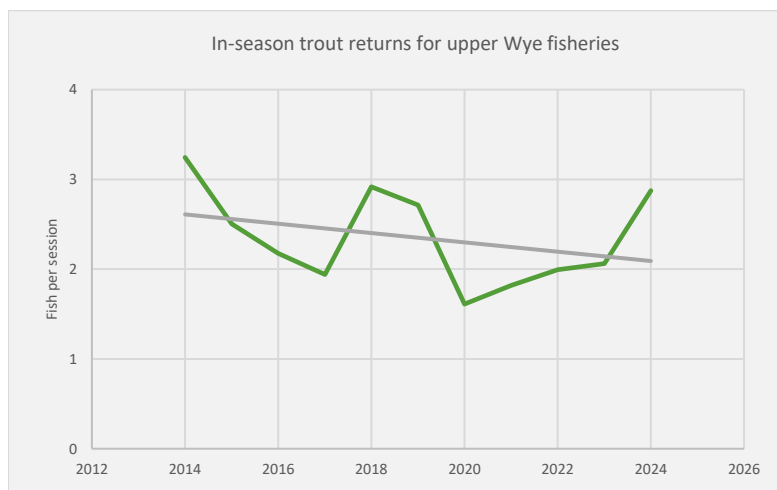


Figure 5: average in-season trout returns for all upper Wye booking office beats. The linear regression line to the data (grey) reveals the underlying trend to be reasonably steady.

Middle Wye

This area covers the main stem from Hay on Wye down to Ross on Wye and catch data is primarily about coarse fish species: barbel, chub and pike (other coarse species are not required data in catch returns). Figure 6 shows average catches for barbel over the last decade.

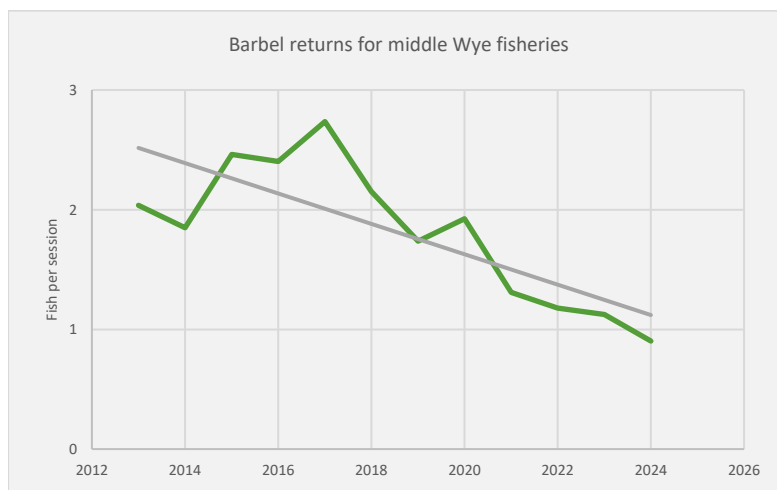


Figure 6: average barbel returns for all middle Wye booking office beats. The linear regression line to the data (grey) reveals the underlying trend to be steadily downwards.

Barbel catches are falling, a trend which appears to have started around 2018. Catches fell precipitously between 2020 and 2021, when the ranunculus disappeared. But rather worryingly, the decline has continued since then, this being suggestive of an ongoing population decline².

It is useful to look at chub catches in the context of barbel catches, the latter being the target species for the majority of anglers. Figure 7 shows average catch returns for both species. Chub catches were fairly steady though 2014-20. Between 2020 and 2021, we see a sudden, two-fold increase in chub captures, largely mirroring the decline in barbel. A reasonable inference would be that the lack of ranunculus cover after 2020 inhibits barbel from venturing into the shallower runs during daylight, leaving only the much bolder chub to find the anglers' bait. Anecdotally, this matches my own experience fishing the middle and lower Wye for barbel since 2008. The extra chub I'm catching are mainly in the 2-5lb class. These are not youngsters and have been in the river a few years, meaning that the increased catch rate is unlikely to be due to population growth.

Other anglers I've spoken to generally remark how poor the barbel fishing is compared to five or six years ago. Such barbel as are being caught on the shallower runs (where the ranunculus used to be) in

² This is not to suggest that these fish have necessarily died (barbel are known to be long range travellers, for example), but many of them are clearly no longer resident in the places where they used to live.

summer are often emerging well after dark (WUF anglers cannot fish after dark). The relatively few swims with suitable structure where barbel feel secure during the daytime are now the only productive areas of river.

Also anecdotally, dace are becoming increasingly rare in the middle river.

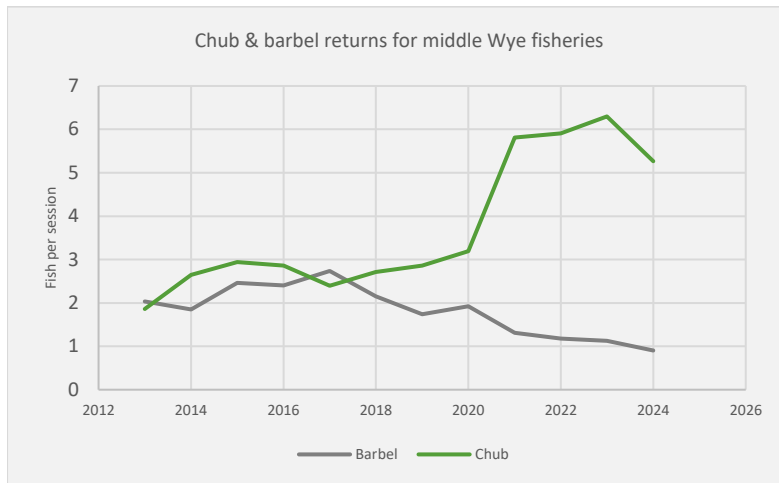


Figure 7: average chub & barbel returns for all middle Wye booking office beats.

Monnow

This area, being what in Victorian times was regarded as one of our top limestone trout rivers, includes two booking office beats, one at Skenfrith and one at Rockfield. Beginning with winter grayling fishing, figure 8 shows an erratic picture with an underlying downward trend. The absolute numbers are similar to the recent Irfon performance (figure 2) and represent very poor fishing for circa £25 per day. Clearly anglers agree, with winter bookings over the last five years averaging less than 20 angler-days shared between the two fisheries.

Over the whole of the 2014-24 period, out of season trout by-catch remained low and steady at just under 1 per angler-day.

In-season trout fishing, however, has been much more erratic as shown in figure 9 and also shows a downward trend. As for grayling, the absolute numbers represent a pretty poor return for time and money invested, at least for the majority of anglers.

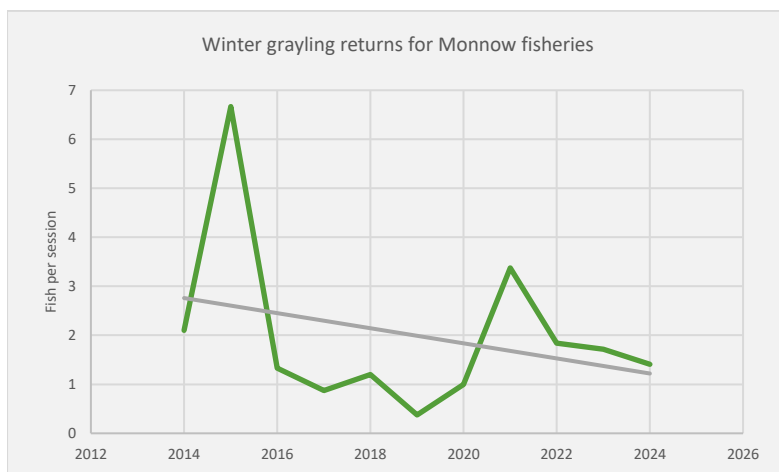


Figure 8: average winter grayling returns for river Monnow booking office beats.

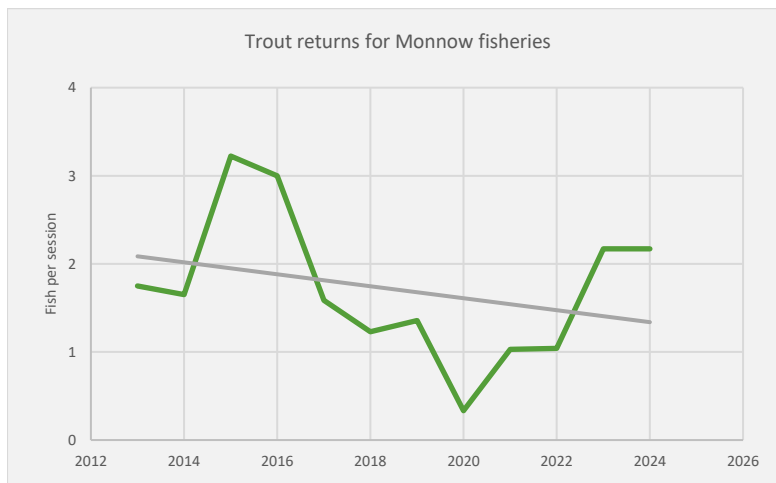


Figure 9: average in-season trout returns for river Monnow booking office beats.

Lugg & Arrow

Filtering for trout and grayling, this area contains three fisheries, one on the Arrow and two on the Lugg upstream of Leominster. The collective averages for winter grayling are shown in figure 10. This shows a markedly healthier situation than we found on the Irfon, Ithon and upper Wye. Whilst there is considerable year to year variability, the general trend over the last decade has been upwards. Some of this may be attributable to the switch away from fly-only in the last few years but these do not appear to be fisheries in decline, providing catch rates not seen on the Irfon since pre-2014.

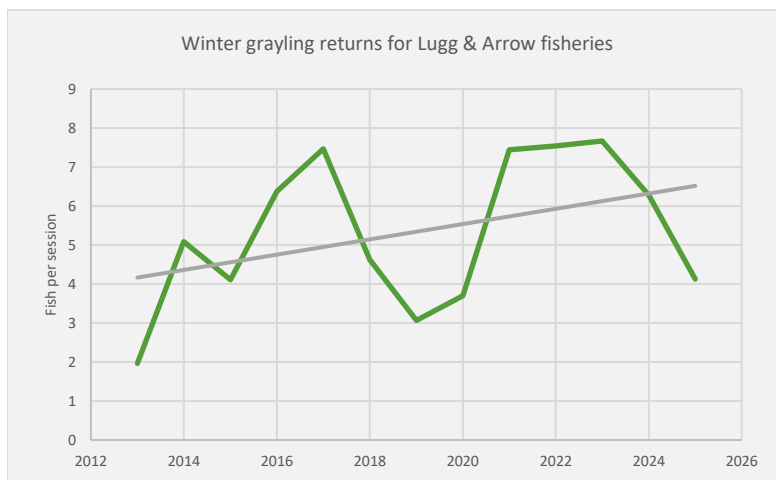


Figure 10: average in-season trout returns for Lugg & Arrow booking office beats.

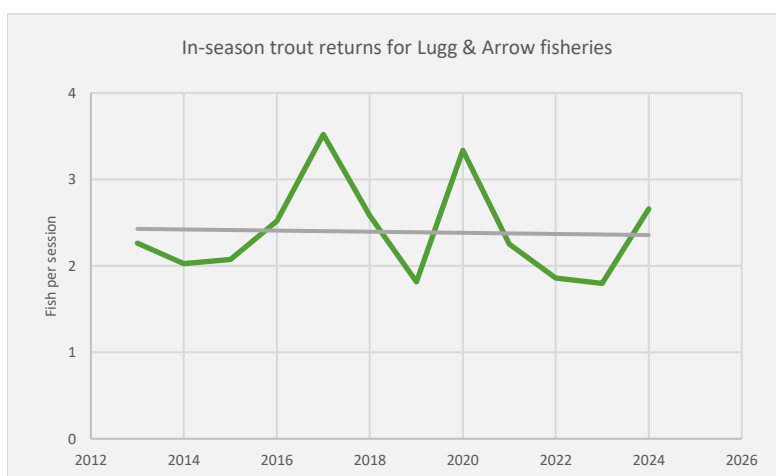


Figure 11: average in-season trout returns for Lugg & Arrow booking office beats.

In-season trout captures have remained steady through the 2014-24 period, as shown by figure 11. Summer grayling captures (fly-only) show the same upward trend seen for winter grayling, suggesting a genuine population increase has occurred.

Discussion

The purpose of this report has been to present and summarise the Fishing Passport catch return data. A discussion of stressors driving change is beyond the scope of this work. Proceeding in this spirit, it is clear that wide areas of the Wye catchment have experienced significant ecological degradation over the period 2014-2024. The only area of consistent trout and grayling catch returns appears to be on the upper Lugg and Arrow, with all other areas showing declining catch rates.

Grayling appear to be in serious decline across the Irfon, Ithon and upper Wye sub catchments, with Irfon & Ithon catches down by a factor of 4 and the main stem by a factor of 2 since 2014. Trout appear to be less impacted by whatever stressors are driving this – at least over the period for which we have data. An important caveat here is to avoid the error of ‘baseline shift’. Ten years is really too short a period to know what things used to be like³. The rapid decline in winter bookings on Cefnllysgwynne, plus my own experiences, suggest that the decline in grayling on the Irfon actually began further back in time, the implication being that the decline is far more than 4-fold. At the current time when salmon numbers are becoming critically low, it is worth pointing out that salmon fry and parr share much of the grayling’s habitat. Stressors on grayling are likely also to be stressors on young salmon. Furthermore, a healthy population of grayling would be expected to provide predator buffering to salmon fry and parr.

In the middle Wye we have found a significant fall in barbel catch rates, including a step change at 2020-21 when the ranunculus disappeared. The continued fall post-2021 is strongly suggestive of a population decline, presumably in response to the massive reduction in available habitat. A sudden rise in chub captures also correlates with the ranunculus extinction and is most likely explained simply by the absence of barbel to compete for anglers’ bait in swims with no macrophyte cover. There is no significant evidence in the data regarding chub populations at present, but it is hard to see how they will not be impacted longer term. Alongside this, we should also consider the anecdotal evidence of falling dace numbers. Additionally, picking up the point about baseline shift, it is worth recalling that prior to the 1990s the middle Wye was home to significant stocks of roach. Furthermore, all of this decline is relevant to salmon. Dace, roach, small chub and other ‘silver fish’ all provide predator buffering to migrating smolts.

The Monnow likewise has declined over the last ten years, both for trout and winter grayling catches. Once again, it is plausible that if we could see comparable data from several decades ago, the changes would be even more marked.

Paradoxically, the Arrow & Lugg area stands out with stable trout catches and increasing grayling catches⁴ over the period examined.

³ Taking salmon catch numbers for example, a 10-year perspective might suggest a fall of circa 2x, but expanding the time horizon back to the 1970s exposes the true fall to be circa 20x.

⁴ Although outside the scope of this report, note that trout returns for the Usk have also been substantially stable over the period examined.

Conclusion

- The Fishing Passport catch returns are a very useful dataset for monitoring fish populations across the Wye catchment. Despite the limited timespan covered to date, it still provides useful trend data and a means of comparing different sub catchments. At a time when regulators' capacity to perform ecological surveys is limited, this dataset is arguably a very useful resource.
- This data should surely be considered alongside electrofishing results when regulators assess the ecological health of catchment water bodies.
- As well as declining salmon numbers, several other fish species in multiple areas of the Wye catchment also appear to be in decline. This needs to be widely understood and urgently addressed within the broader attempts to restore the Wye catchment to health.

Acknowledgements

Thanks are due to WUF for, albeit unknowingly, providing the data.